



Algorithm Design IV

Divide and Conquer I

Guoqiang Li
School of Computer Science



SHANGHAI JIAO TONG
UNIVERSITY



The **divide-and-conquer** strategy solves a problem by:

- 1 Breaking it into **subproblems** that are themselves smaller instances of the same type of problem.
- 2 **Recursively** solving these subproblems.
- 3 **Appropriately** combining their answers.

Multiplication

Product of Complex Numbers



Carl Friedrich Gauss(1777-1855) noticed that although the product of two complex numbers

$$(a + bi)(c + di) = ac - bd + (bc + ad)i$$

seems to involve **four** real-number multiplications, it can in fact be done with just **three**: ac , bd , and $(a + b)(c + d)$, since

$$bc + ad = (a + b)(c + d) - ac - bd$$

- In big O way of thinking, reducing the number of multiplications from **four** to **three** seems wasted ingenuity.
- But this modest improvement becomes very significant when applied **recursively**.

Multiplication



Suppose x and y are two n -integers, and assume for convenience that n is a power of 2.

[Hints: For every n there exists an n' with $n \leq n' \leq 2n$ such that n' a power of 2.]

As a first step toward multiplying x and y , we split each of them into their left and right halves, which are $n/2$ bits long

$$x = \boxed{x_L} \boxed{x_R} = 2^{n/2}x_L + x_R$$

$$y = \boxed{y_L} \boxed{y_R} = 2^{n/2}y_L + y_R$$

$$xy = (2^{n/2}x_L + x_R)(2^{n/2}y_L + y_R) = 2^n x_L y_L + 2^{n/2}(x_L y_R + x_R y_L) + x_R y_R$$

Additions and multiplications by powers of 2 take linear time.

Multiplication



The additions take linear time, as do multiplications by powers of 2 (that is, $O(n)$).

The significant operations are the four $n/2$ -bit **multiplications**: these can be handled by four **recursive calls**.

Writing $T(n)$ for the overall running time on n -bit inputs, we get the **recurrence relations**:

$$T(n) = 4T(n/2) + O(n)$$

Solution: $O(n^2)$

By **Gauss's** trick, three multiplications $x_L y_L$, $x_R y_R$, and $(x_L + x_R)(y_L + y_R)$ suffice.



Alternative: Karatsuba's trick

To multiply two n -bit integers x and y :

- Divide x and y into low- and high-order bits.
- To compute middle term $bc + ad$, use identity:

$$bc + ad = ac + bd - (a - b)(c - d)$$

- Also multiply only three $n/2$ -bit integers, recursively.

Conjecture. [Kolmogorov 1956] Grade-school algorithm is optimal.

Theorem. [Karatsuba 1960] Conjecture is false.

Algorithm for Integer Multiplication



MULTIPLY (x, y)

Two positive integers x and y , in binary;

$n = \max(\text{size of } x, \text{size of } y)$ rounded as a power of 2;

if $n = 1$ **then** return (xy);

$x_L, x_R =$ leftmost $n/2$, rightmost $n/2$ bits of x ;

$y_L, y_R =$ leftmost $n/2$, rightmost $n/2$ bits of y ;

$P_1 = \text{MULTIPLY}(x_L, y_L)$;

$P_2 = \text{MULTIPLY}(x_R, y_R)$;

$P_3 = \text{MULTIPLY}(x_L + x_R, y_L + y_R)$;

return ($P_1 \times 2^n + (P_3 - P_1 - P_2) \times 2^{n/2} + P_2$)



The recurrence relation

$$T(n) = 3T(n/2) + O(n)$$

The algorithm's recursive calls form a **tree structure**.

At each successive level of recursion the subproblems get **halved** in size.

At the $(\log_2 n)^{th}$ level, the subproblems get down to size **1**, and so the recursion ends.

The **height** of the tree is $\log_2 n$.

The **branch factor** is **3**: each problem produces three smaller ones, with the result that at depth k there are 3^k subproblems, each of **size** $n/2^k$.

For each subproblem, a **linear** amount of work is done in combining their answers.

Time Analysis



The total time spent at depth k in the tree is

$$3^k \times O\left(\frac{n}{2^k}\right) = \left(\frac{3}{2}\right)^k \times O(n)$$

At the top level, when $k = 0$, we need $O(n)$.

At the bottom, when $k = \log_2 n$, it is $O(3^{\log_2 n}) = O(n^{\log_2 3})$

The work done increases **geometrically** from $O(n)$ to $O(n^{\log_2 3})$, by a factor of $3/2$ per level.

The sum of any **increasing geometric series** is, within a constant factor, the last term of the series.

Therefore, the overall running time is

$$O(n^{\log_2 3}) \approx O(n^{1.59})$$

Time Analysis



SHANGHAI JIAO TONG
UNIVERSITY

Q: Can we do better?

- Yes!

Recurrence Relations



Master Theorem

If $T(n) = aT(\lceil n/b \rceil) + O(n^d)$ for some constants $a > 0$, $b > 1$ and $d \geq 0$, then

$$T(n) = \begin{cases} O(n^d) & \text{if } d > \log_b a \\ O(n^d \log n) & \text{if } d = \log_b a \\ O(n^{\log_b a}) & \text{if } d < \log_b a \end{cases}$$

The Proof of the Theorem



Proof:

Assume that n is a power of b .

The size of the subproblems decreases by a factor of b with each level of recursion, and therefore reaches the base case after $\log_b n$ levels - the height of the recursion tree.

Its branching factor is a , so the k -th level of the tree is made up of a^k subproblems, each of size n/b^k .

$$a^k \times O\left(\frac{n}{b^k}\right)^d = O(n^d) \times \left(\frac{a}{b^d}\right)^k$$

k goes from 0 to $\log_b n$, these numbers form a geometric series with ratio a/b^d , comes down to three cases.

The Proof of the Theorem



The ratio is less than 1.

Then the series is decreasing, and its sum is just given by its first term, $O(n^d)$.

The ratio is greater than 1.

The series is increasing and its sum is given by its last term, $O(n^{\log_b a})$

The ratio is exactly 1.

In this case all $O(\log n)$ terms of the series are equal to $O(n^d)$.



Theorem (Akra–Bazzi 1998)

Given constants $a_i > 0$ and $0 < b_i < 1$, functions $|h_i(n)| = O(n/\log^2 n)$ and $g(n) = O(n^c)$. If $T(n)$ satisfies the recurrence:

$$T(n) = \sum_{i=1}^k a_i T(b_i n + h_i(n)) + g(n)$$

then, $T(n) = \Theta\left(n^p \left(1 + \int_1^n \frac{g(u)}{u^{p+1}} du\right)\right)$, where p satisfies $\sum_{i=1}^k a_i b_i^p = 1$.

Example. $T(n) = T(\lfloor n/5 \rfloor) + T(n - 3\lfloor n/10 \rfloor) + 11/5n$, with $T(0) = 0$ and $T(1) = 0$.

- $a_1 = 1, b_1 = 1/5, a_2 = 1, b_2 = 7/10 \Rightarrow p = 0.83978 \dots < 1$.
- $h_1(n) = \lfloor n/5 \rfloor - n/5, h_2(n) = 3/10n - 3\lfloor n/10 \rfloor$.
- $g(n) = 11/5n \Rightarrow T(n) = \Theta(n)$.

Quiz



SHANGHAI JIAO TONG
UNIVERSITY

$$T(n) = 4 \cdot T(\sqrt{n}) + 1$$

Merge Sort

The Algorithm



```
MERGESORT ( $a[1 \dots n]$ )  
An array of numbers  $a[1 \dots n]$ ;  
if  $n > 1$  then  
    | return (MERGE (MERGESORT ( $a[1 \dots \lfloor n/2 \rfloor]$ ),  
    |   MERGESORT ( $a[\lfloor n/2 \rfloor + 1 \dots n]$ )) )  
end  
else return ( $a$ );
```

```
MERGE ( $x[1 \dots k], y[1 \dots l]$ )  
if  $k = 0$  then return  $y[1 \dots l]$ ;  
if  $l = 0$  then return  $x[1 \dots k]$ ;  
if  $x[1] \leq y[1]$  then return (  $x[1] \circ \text{MERGE} (x[2 \dots k], y[1 \dots l])$  ) ;  
else return (  $y[1] \circ \text{MERGE} (x[1 \dots k], y[2 \dots l])$  ) ;
```

An Iterative Version



```
ITERATIVE-MERGESORT( $a[1 \dots n]$ )  
An array of numbers  $a[1 \dots n]$ ;  
 $Q = [ ]$  empty queue;  
for  $i = 1$  to  $n$  do  
|   Inject( $Q, [a[i]]$ );  
end  
while  $|Q| > 1$  do  
|   Inject( $Q, \text{MERGE}(\text{Eject}(Q), \text{Eject}(Q))$ );  
end  
return( $\text{Eject}(Q)$ );
```

The Time Analysis



SHANGHAI JIAO TONG
UNIVERSITY

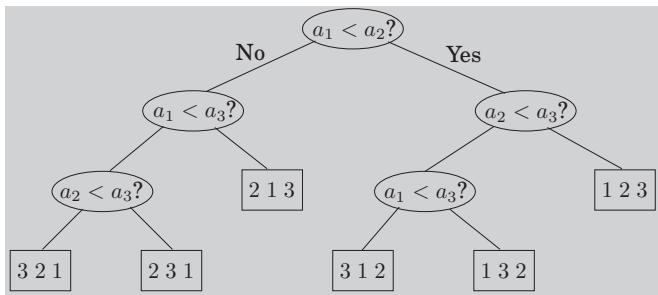
The recurrence relation:

$$T(n) = 2T(n/2) + O(n)$$

By Master Theorem:

$$T(n) = O(n \log n)$$

Q: Can we do better?

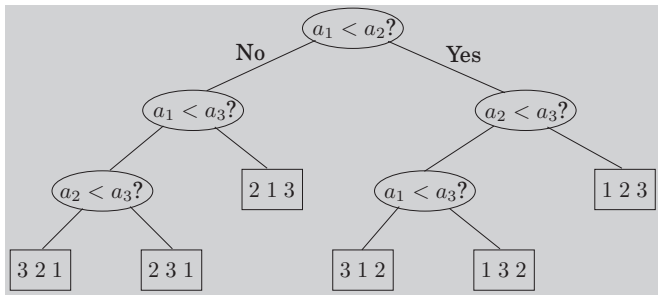


Sorting algorithms can be depicted as **trees**.

The **depth** of the tree - the number of comparisons on the longest path from root to leaf, is the worst-case time complexity of the algorithm.

Assume n elements. Each of its leaves is labeled by a **permutation** of $\{1, 2, \dots, n\}$.

Sorting



Every permutation must appear as the label of a leaf.

This is a binary tree with $n!$ leaves.

So, the depth of the tree - and the complexity of the algorithm - must be at least

$$\log(n!) \approx \log(\sqrt{\pi(2n+1/3)} \cdot n^n \cdot e^{-n}) = \Omega(n \log n)$$

Median



The **median** of a list of numbers is its **50th** percentile: half the number are bigger than it, and half are smaller.

If the list has **even** length, we pick the smaller one of the two.

The purpose of the median is to summarize a set of numbers by a single typical value.

Computing the median of n numbers is easy, just sort them. ($O(n \log n)$).

Q: Can we do better?

Selection



Input: A list of number S ; an integer k .

Output: The k th smallest element of S .

A Randomized Selection



For any number v , imagine splitting list S into three categories:

- elements smaller than v , i.e., S_L ;
- those equal to v , i.e., S_v (there might be duplicates);
- and those greater than v , i.e., S_R ; respectively.

$$selection(S, k) = \begin{cases} selection(S_L, k) & \text{if } k \leq |S_L| \\ v & \text{if } |S_L| < k \leq |S_L| + |S_v| \\ selection(S_R, k - |S_L| - |S_v|) & \text{if } k > |S_L| + |S_v| \end{cases}$$

How to Choose v ?



It should be picked quickly, and it should shrink the array substantially, the ideal situation being

$$|S_L|, |S_R| \approx \frac{|S|}{2}$$

If we could always guarantee this situation, we would get a running time of

$$T(n) = T(n/2) + O(n) = O(n)$$

But this requires picking v to be the median, which is our ultimate goal!

Instead, we pick v randomly from S !

How to Choose v ?



Worst-case scenario would force our selection algorithm to perform

$$n + (n - 1) + (n - 2) + \dots + \frac{n}{2} = \Theta(n^2)$$

Best-case scenario $O(n)$

The Efficiency Analysis



v is **good** if it lies within the 25th to 75th percentile of the array that it is chosen from.

A randomly chosen v has a 50% chance of being good.

Lemma

*On average a fair coin needs to be tossed two times before a **heads** is seen.*

Proof:

Let E be the expected number of tosses before heads is seen.

$$E = 1 + \frac{1}{2}E$$

Therefore, $E = 2$.

The Efficiency Analysis



SHANGHAI JIAO TONG
UNIVERSITY

Let $T(n)$ be the expected running time on the array of size n , we get

$$T(n) \leq T(3n/4) + O(n) = O(n)$$

Matrix Multiplication



The product of two $n \times n$ matrices X and Y is a $n \times n$ matrix $Z = XY$, with which $(i, j)th$ entry

$$Z_{ij} = \sum_{k=1}^n X_{ik} Y_{kj}$$

In general, matrix multiplication is not commutative, say, $XY \neq YX$

The running time for matrix multiplication is $O(n^3)$

- There are n^2 entries to be computed, and each takes $O(n)$ time.



Matrix multiplication can be performed **blockwise**.

$$X = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \quad Y = \begin{bmatrix} E & F \\ G & H \end{bmatrix}$$

$$XY = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} E & F \\ G & H \end{bmatrix} = \begin{bmatrix} AE + BG & AF + BH \\ CE + DG & CF + DH \end{bmatrix}$$

$$T(n) = 8T(n/2) + O(n^2)$$

$$T(n) = O(n^3)$$



$$X = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \quad Y = \begin{bmatrix} E & F \\ G & H \end{bmatrix}$$

$$XY = \begin{bmatrix} P_5 + P_4 - P_2 + P_6 & P_1 + P_2 \\ P_3 + P_4 & P_1 + P_5 - P_3 - P_7 \end{bmatrix}$$

$$P_1 = A(F - H) \quad P_5 = (A + D)(E + H)$$

$$P_2 = (A + B)H \quad P_6 = (B - D)(G + H)$$

$$P_3 = (C + D)E \quad P_7 = (A - C)(E + F)$$

$$P_4 = D(G - E)$$

$$T(n) = 7T(n/2) + O(n^2)$$

$$T(n) = O(n^{\log_2 7}) \approx O(n^{2.81})$$